LF1. Machine-Verified Volume Typicality and Frequency Convergence for Deterministic Repeated Trials

A Preprint

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Abstract

We formalise in Lean4/Mathlib the core repeated-trial theorem underlying volume-based deterministic accounts of quantum statistics. The setting is a finite-measure ontic state space (Σ, μ_L) equipped with a measurable preparation region $\Omega_0 \subseteq \Sigma$, a deterministic measurable flow $\Phi : \Sigma \to \Sigma$, and a fixed measurable ontic outcome region. The key modelling assumption is explicit: repeated trials are represented by a repeated-preparation model in which each trial begins from an initial microstate distributed according to the conditional preparation measure on Ω_0. For a fixed outcome region, the empirical frequencies of the associated indicator observables converge almost surely to the corresponding normalised ontic weight.

The formalisation isolates the exact logical dependencies of the result, separates what is proved inside Lean from what is imported from Mathlib, and distinguishes the physical assumptions of the repeated-preparation model from the law-of-large-numbers machinery used in the convergence step. In the current repository, the theorem is exported as `LF1_main_theorem_ae`. It is stated for a repeated-trial model $T : S.\mathrm{TrialModel}\,\Omega$ and a fixed outcome region $O : S.\mathrm{OutcomeRegion}$, with pairwise independence of the associated indicator random variables as the only non-trivial repeated-trial hypothesis supplied explicitly at the final stage. Integrability and identical-distribution properties are discharged internally from the trial-model structure.

Within the Constraint-Surface Dynamics programme, this paper formalises the statistical backbone of volume typicality in the notation of TN0. It does not, by itself, provide a machine-verified derivation of Born-form weights, a projective measure bridge, or the mixed-state and post-measurement layers. Its role is narrower and more precise: to show that, once deterministic ontic events and a repeated-preparation scheme have been specified, empirical frequencies converge almost surely to the corresponding real-valued normalised ontic weights.

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1 Introduction

1.1 Why formalise the volume-typicality backbone

A central claim of volume-based deterministic approaches to quantum statistics is that stable experimental frequencies can arise from deterministic dynamics together with geometric measure, without introducing fundamental stochastic laws at the ontic level. In the Constraint-Surface Dynamics (CSD) programme, this claim appears in its most basic form as a repeated-trial theorem: if an experiment is represented by a measurable preparation region and a measurable partition into outcome regions, and if each run begins from an independently sampled initial microstate drawn from the preparation ensemble, then long-run outcome frequencies converge to the corresponding normalised volume ratios (Blore 2026a).

This theorem is mathematically standard in spirit, but its role in foundational work is often presented only informally. In practice, several distinct ingredients are usually run together: the deterministic dynamical law, the preserved measure, the preparation model, the measurable outcome partition, and the probabilistic convergence theorem used to pass from single-run geometry to long-run frequencies. A machine-checked formalisation is valuable precisely because it forces these ingredients to be stated separately and with full logical precision. Assumptions that may remain tacit in an informal derivation must be declared explicitly. Dependencies on external theorems must be identified. Ambiguities of notation or scope must be removed.

The purpose of the present paper is therefore not to introduce a new physical postulate, but to formalise a specific theorem that already plays a foundational role in the CSD programme. We use Lean4 and Mathlib to make the repeated-trial frequency claim fully explicit, to separate the mathematical core from the physical modelling assumptions, and to provide a reusable formal basis for later work on the projective measure bridge, operational weights, mixed states, control Hamiltonians, and sequential update.

1.2 Scope of the present paper

The purpose of LF1 is to formalise, in Lean4/Mathlib, the repeated-trial frequency theorem that underlies the volume-typicality layer of the CSD programme. The formal result is stated at the level of an abstract ontic measurable space equipped with a finite measure, a measurable preparation region, a deterministic measurable flow, and a fixed measurable ontic outcome region. The theorem then concerns repeated trials whose initial microstates are sampled from the conditional preparation measure.

The present paper therefore does not formalise a full measurement partition as a primitive Lean object. Instead, it formalises the theorem for a single measurable outcome region at a time. In physical applications, a measurement context may of course determine a family of such regions, typically forming a partition up to null sets, but the Lean theorem is deliberately stated one region at a time in order to isolate the exact repeated-trial content.

Equally importantly, LF1 does not attempt to formalise the measure bridge to projective space, the Born-form weight assignment, mixed states, POVMs, subsystem reduction, or post-measurement update. Those belong to later layers of the programme. LF1 isolates the statistical backbone only: once a deterministic ontic model and a repeated-preparation scheme have been specified, empirical frequencies converge almost surely to the corresponding normalised ontic weight.

1.3 Main result

The corresponding Lean4 implementation is publicly available at `https://github.com/zblore/csd-lean4`. This repository contains the LF1 development and serves as the implementation base for the theorem `LF1_main_theorem_ae`.

It is stated for a repeated-trial model $T : S.\mathrm{TrialModel}\,\Omega$, a fixed measurable ontic outcome region $O : S.\mathrm{OutcomeRegion}$, and a pairwise-independence hypothesis on the associated indicator random variables. Its conclusion is almost-sure convergence of the empirical frequencies to the real-valued ontic weight $O.\mathrm{weightReal}$.

In mathematical terms, the theorem has the form

$$\hat{w}_N(\omega) \longrightarrow O.\mathrm{weightReal} \qquad \text{for } T.\mathrm{trialMeasure}\text{-almost every } \omega,$$

where $\hat{w}_N$ is the empirical frequency built from the indicator observable associated with the outcome region O. The key point is that the theorem is deterministic at the ontic level. A single trial has no stochastic outcome law. The only probabilistic ingredient is the repeated-preparation model on the initial conditions.

The theorem is intentionally modest in scope. It does not prove a Born-rule statement. It proves that, once the ontic model, preparation measure, and outcome event are fixed, repeated preparation yields frequencies converging almost surely to the corresponding normalised ontic weight. This is the precise machine-verified content of LF1.

1.4 Why machine verification matters here

One contribution of the present work is methodological. Foundational arguments often derive much of their apparent force from compression. Several steps are packaged together and presented as conceptually natural, even when their exact logical order has not been made fully explicit. Formal verification is valuable because it resists that compression. To formalise the repeated-trial theorem, one must answer concrete questions such as the following.

1. What exactly is the ambient measurable structure on Σ.
2. Which subsets are assumed measurable.
3. How is the conditional preparation measure defined.
4. Where independence enters the repeated-trial model.
5. Which parts of the proof are internal to the theorem prover and which rely on imported probability theory.
6. Which assumptions belong to mathematics and which belong to the physical model.

For a sceptical reader, this clarification is itself a substantive result. Even if the formalised theorem is mathematically familiar, the act of machine verification exposes the precise assumptions required for the theorem to carry the physical interpretation assigned to it. This is especially important in foundational settings, where claims may otherwise appear stronger than the formal argument strictly warrants.

1.5 Position within the CSD programme

Within the wider CSD programme, LF1 occupies a specific and limited role. It formalises the repeated-trial volume-typicality layer that underpins the statistical claims of Paper A, using the notation and symbol conventions fixed in TN0 (Blore 2026a; Blore 2026d). Later layers of the programme address different questions. The measure-bridge layer connects ontic Liouville volume to projective Fubini-Study measure. The operational layer treats mixed states, POVMs, and subsystem reduction. The control layer treats control-dependent Hamiltonians and outcome basins. The sequential layer treats outcome-conditioned update and repeated circuits. LF1 is the first element of that chain because it isolates the cleanest mathematical kernel and the theorem most naturally suited to immediate Lean formalisation.

This limited role should be kept in view throughout. LF1 is important because later claims depend on it. It is not, by itself, the whole finite-dimensional story.

1.6 Structure of the paper

The remainder of the paper is organised as follows.

Section 2 states the mathematical theorem in the notation used throughout the current CSD programme and defines the repeated-trial preparation model precisely.

Section 3 presents the Lean4 formalisation architecture, including the measure-theoretic setup, the trial space, the indicator observables, and the convergence argument.

Section 4 provides an assumptions audit, separating internally proved lemmas, Mathlib imports, and physical modelling assumptions.

Section 5 explains how LF1 fits into the wider CSD programme and which later formalisation layers depend on it.

Section 6 discusses what the formalisation clarifies, what remains outside its scope, and what the natural next steps are for the Lean branch.

2 Mathematical statement of the repeated-trial theorem

2.1 Ontic setup

Let $(\Sigma, \mathcal{F}, \mu_L)$ be a finite-measure ontic measurable space. Let

$$\Phi : \Sigma \to \Sigma$$

be a measurable deterministic evolution map, interpreted physically as the ontic flow associated with a fixed experimental context. Let

$$\Omega_0 \subseteq \Sigma$$

be the measurable preparation region, with

$$0 < \mu_L(\Omega_0) < \infty.$$

For the formal LF1 theorem, one next fixes a single measurable ontic outcome region

$$\Omega^\Sigma \subseteq \Sigma.$$

This region is to be understood as the ontic basin associated with one definite outcome under the chosen experimental context. The corresponding pulled-back initial-condition event is

$$\Phi^{-1}(\Omega^\Sigma),$$

and the corresponding preparation-side event is

$$\Omega_0 \cap \Phi^{-1}(\Omega^\Sigma).$$

This single-region formulation is the exact level at which the current Lean theorem is stated. In physical applications, a measurement context may determine a family of outcome regions $\{\Omega_i^\Sigma\}_{i \in I}$, often forming a measurable partition up to null sets. However, LF1 formalises the theorem one outcome region at a time, which is sufficient for the repeated-trial convergence result.

2.2 Conditional preparation measure

The file `Preparation.lean` constructs the conditional preparation measure from the ontic measure μ_L and the preparation region Ω_0. At the mathematical level, the construction is

$$\mu_{\mathrm{prep}}(A) := \frac{\mu_L(A \cap \Omega_0)}{\mu_L(\Omega_0)}.$$

In the current repository this is not merely planned architecture. It is already implemented, together with the explicit rewriting lemma

`prepMeasure_apply`,

which expresses the preparation measure in the concrete form above. This lemma is important because it gives the bridge between the abstract probability measure used in the repeated-trial theorem and the Liouville-volume interpretation required by the wider CSD programme.

Accordingly, LF1 does not merely assert the existence of a normalised preparation measure. It already machine-verifies the basic measure-theoretic infrastructure needed to use that measure explicitly in later Lean layers.

2.3 Repeated trials and empirical frequencies

The repeated-trial model is conceptually the usual repeated-preparation scheme: each run begins from a fresh initial microstate sampled from the conditional preparation measure on Ω_0, and each trial then evolves deterministically under Φ. In the current Lean development, however, this is encoded in a slightly more general way.

Rather than building the theorem directly on a specific countable product probability space, the formalisation introduces a repeated-trial structure

$$T : S.\mathrm{TrialModel}\,\Omega,$$

where Ω is an external sample space indexing repeated runs. The structure carries a probability measure P on Ω and measurable trial maps

$$X_n : \Omega \to \Sigma, \qquad n \in \mathbb{N},$$

such that each X_n has law equal to the conditional preparation measure μ_{prep}.

For a fixed ontic outcome region Ω^{Σ}, the event that the n-th trial realises that outcome is the set

$$E_n := X_n^{-1}\big(\Phi^{-1}(\Omega^{\Sigma})\big).$$

Equivalently, writing O for the fixed outcome region object in the Lean development, one has

$$E_n = X_n^{-1}(O.\mathrm{preEvent}).$$

The associated indicator observable is then

$$Y_n := \mathbf{1}_{E_n},$$

and the empirical frequency after N trials is

$$\hat{w}_N := \frac{1}{N}\sum_{n=0}^{N-1} Y_n.$$

This is the quantity whose almost-sure convergence is established in the formal theorem.

2.4 Main theorem

We now state the theorem in the form that matches the current Lean repository.

Theorem 1 (LF1 repeated-trial convergence theorem)**.** *Let S be an ontic setup, let*

$$T : S.\mathrm{TrialModel}\,\Omega$$

be a repeated-trial model, and let

$$O : S.\mathrm{OutcomeRegion}$$

be a fixed measurable ontic outcome region. Assume that the associated indicator random variables are pairwise independent. Then the empirical frequency of the outcome region O converges almost surely to the corresponding real-valued ontic weight:

$$\hat{w}_N \longrightarrow O.\mathrm{weightReal} \qquad \textit{for } T.\mathrm{trialMeasure}\textit{-almost every repeated-trial sample point.}$$

Here the quantity $O.\mathrm{weightReal}$ is the real-valued normalised ontic weight associated with the preparation-side event

$$\Omega_0 \cap \Phi^{-1}(\Omega^{\Sigma}).$$

Equivalently, it is the real-valued form of the preparation probability of the pulled-back outcome event.

The only non-trivial repeated-trial hypothesis supplied explicitly at this stage is pairwise independence of the indicator random variables. Integrability and identical distribution are automatic consequences of the model structure already encoded in the earlier Lean files. This exact separation of assumptions is one of the main clarifications provided by the formalisation.

2.5 Interpretation of the theorem

The theorem does not state that a single trajectory explores the full preparation region in time, nor does it rely on an ergodic hypothesis for individual ontic histories. Instead, it concerns repeated preparation. The statistical regularity arises because each trial begins from a fresh initial microstate sampled from the same preparation ensemble, and because the outcome rule is represented by measurable basin membership after deterministic evolution.

This distinction is important. The theorem formalised here is a statement about deterministic repeated trials with ensemble sampling, not about arbitrary long-time averages along one ontic orbit. In particular, the probabilistic content enters through the repeated-trial preparation model, not through any stochastic modification of the ontic dynamics.

2.6 Boundary of the claim

Theorem 1 establishes a frequency result for deterministic measure-preserving dynamics under repeated preparation. It provides the statistical backbone required for volume-typicality arguments, but it does not by itself identify the weights w_i with Born-form probabilities on projective state space.

Two additional layers lie beyond the scope of the present paper.

First, one must connect the relevant ontic volume structure to the effective projective measure used in the quantum description. Second, one must justify the operational probability rule on that projective space. Those steps belong to later parts of the CSD programme and to later papers in the Lean formalisation branch.

Accordingly, LF1 should be read as a formalisation of the determinism-to-frequency bridge only. It is a necessary component of the wider programme, but not its complete probability-theoretic closure.

3 Lean4 formalisation architecture

3.1 Overview

The formal content of LF1 is organised around a simple theorem chain. We begin with a finite-measure ontic state space, a measurable preparation region, and a fixed measurable ontic outcome region. We then define the repeated-trial preparation model by equipping an external sample space with trial maps whose common law is the conditional preparation measure on the preparation region. From these sampled initial conditions, we construct indicator random variables for the chosen outcome region and form the corresponding empirical frequencies. The main theorem then follows by combining three ingredients: measurability of the indicator observables, identification of their expectations with the corresponding normalised ontic weight, and an appropriate law of large numbers.

This section gives a guided description of that architecture. It is not intended as a code dump. Instead, the goal is to explain how the mathematical statement of Section 2 is decomposed into a sequence of formal definitions and lemmas suitable for Lean4 and Mathlib. The main design principle is modularity. The formalisation isolates the purely measure-theoretic kernel of the repeated-trial theorem so that later papers in the Lean branch can reuse it without reworking the foundational infrastructure. In physical applications, a full measurement context may determine a family of outcome regions, but the present formal theorem is stated one region at a time.

3.2 Base measurable structure

The base file `Setup.lean` introduces the ambient ontic data for LF1. Its core structure is an abstract measurable space Σ equipped with:

1. a finite measure μ_L,
2. a deterministic evolution map $\Phi : \Sigma \to \Sigma$,
3. a measure-preservation hypothesis for Φ,
4. a measurable preparation region $\Omega_0 \subseteq \Sigma$,
5. a non-vanishing assumption $\mu_L(\Omega_0) \neq 0$.

A noteworthy feature of the current implementation is that the measure-preservation hypothesis is carried as part of the ontic model structure, but the LF1 proofs themselves directly use only measurability of Φ, extracted from that hypothesis. This is the correct division for the present layer: LF1 needs the measurable pullback of outcome events, while the fuller dynamical significance of Liouville preservation is reserved for later layers.

Outcome regions are not fields of the base setup structure. They are introduced later, in `Outcomes.lean`, as separate measurable objects associated with a fixed experimental context.

3.3 Conditional preparation measure

The file `Preparation.lean` constructs the preparation probability measure induced by the preparation region Ω_0. Mathematically, this is the normalised restriction of μ_L to Ω_0:

$$\mu_{\mathrm{prep}}(A) = \frac{\mu_L(A \cap \Omega_0)}{\mu_L(\Omega_0)}.$$

In the current repository this construction is already implemented, together with the explicit rewriting lemma

`prepMeasure_apply`,

which expresses the preparation measure in the concrete form above. This step is conceptually modest but formally important. It isolates the point at which the physical preparation procedure becomes a probability measure suitable for repeated sampling.

The file also establishes the basic facts needed downstream: the conditional preparation measure is well defined because

$$0 < \mu_L(\Omega_0) < \infty,$$

it is a probability measure, and for every measurable $A \subseteq \Sigma$ one has the explicit identity

$$\mu_{\mathrm{prep}}(A) = \frac{\mu_L(A \cap \Omega_0)}{\mu_L(\Omega_0)}.$$

This infrastructure matters because it cleanly separates the ontic measure μ_L, which belongs to the physical model, from the preparation probability law, which is the object used by the repeated-trial probability theory. It also provides a reusable measure-theoretic layer for later Lean papers in the branch.

3.4 Outcome events and weights

The file `Outcomes.lean` introduces a single measurable ontic outcome region

$$O : S.\mathrm{OutcomeRegion}.$$

Its underlying measurable set is denoted $O.\Omega$. From this the formalisation defines three associated objects.

First, the pulled-back outcome event under the deterministic ontic flow:

$$O.\mathrm{preEvent} := \Phi^{-1}(O.\Omega).$$

Second, the preparation-side event:

$$O.\mathrm{prepEvent} := \Omega_0 \cap O.\mathrm{preEvent}.$$

Third, the corresponding outcome weight under the preparation measure:

$$O.\mathrm{weight} := \mu_{\mathrm{prep}}(O.\mathrm{preEvent}), \qquad O.\mathrm{weightReal} := \mathrm{toReal}\big(O.\mathrm{weight}\big).$$

A central lemma already present in the repository states that

$$O.\mathrm{weight} = \frac{\mu_L(O.\mathrm{prepEvent})}{\mu_L(\Omega_0)}.$$

This makes the volume-typicality interpretation explicit: the ontic weight is the fraction of the preparation region whose deterministic evolution lands in the chosen outcome region.

The file `Expectation.lean` then proves the expectation bridge in two steps. First,

$$\mathbb{E}[Y_n] = \texttt{trialProbReal}(O, n),$$

via the lemma `integral_indicatorRV_eq_trialProbReal`. Second,

$$\mathbb{E}[Y_n] = O.\mathrm{weightReal},$$

via the lemma `integral_indicatorRV_eq_weightReal`. These are the exact machine-verified identities that connect indicator expectations to normalised ontic weights.

3.5 Repeated-trial probability space

A natural informal presentation of repeated trials uses a countable product probability space and coordinate projections. The current repository adopts a slightly more abstract and more reusable interface. The file `Trials.lean` introduces a structure

$$T : S.\mathrm{TrialModel}\,\Omega,$$

where Ω is an external sample space for repeated runs, P is a probability measure on Ω, and

$$X_n : \Omega \to \Sigma$$

is the ontic initial microstate for the n-th trial.

Each X_n is measurable, and each has law equal to the preparation probability measure:

$$\mathrm{Law}(X_n) = \mu_{\mathrm{prep}}.$$

This is built directly into the structure. Independence is not hard-coded into `TrialModel`. Instead, it is supplied later, at the convergence stage, in the exact form required by the law of large numbers used in the repository.

For a fixed outcome region O, the n-th trial event is

$$T.\mathrm{trialEvent}(O, n) = X_n^{-1}(O.\mathrm{preEvent}),$$

and its probability is

$$T.\mathrm{trialProb}(O, n).$$

The formalisation proves that

$$T.\mathrm{trialProb}(O, n) = O.\mathrm{weight},$$

so every trial carries the same ontic weight because every trial has the same preparation law.

This architecture is preferable for LF1 because it isolates the truly necessary repeated-trial content without committing the theorem statement to one specific concrete sample-space construction.

3.6 Indicator observables and empirical frequencies

For a fixed outcome region O, the formalisation defines the real-valued indicator random variable

$$Y_n := \mathbf{1}_{T.\mathrm{trialEvent}(O,n)}$$

on the repeated-trial sample space. The empirical frequency is then defined by

$$\hat{w}_N := \frac{1}{N} \sum_{n=0}^{N-1} Y_n.$$

At this stage the code proves the standard elementary properties required for the strong-law step. Each indicator is measurable, bounded, and integrable. Moreover, because each trial random variable X_n has the same law μ_{prep}, the indicator family is identically distributed in the sense needed for the later convergence theorem.

It is important, however, to match the repository precisely here. The final theorem does not ask the user to supply a full i.i.d. package as a separate hypothesis. Identical distribution is derived automatically from the `TrialModel` structure and the expectation bridge proved in the earlier files. The only non-trivial repeated-trial hypothesis that must be supplied explicitly at the final stage is pairwise independence of the indicator random variables.

3.7 Convergence theorem

The file `Convergence.lean` applies the strong law of large numbers to the indicator observables attached to a fixed outcome region. The imported Mathlib result used here is

`strong_law_ae_real`.

The first convergence lemma proved in the repository is

`strongLaw_indicator_to_mean_ae`,

which states that the empirical averages of the indicator random variables converge almost surely to their common expectation. The next lemma,

`strongLaw_indicator_to_weight_ae`,

combines that strong-law step with the already-proved expectation-to-weight bridge. Finally,

`strongLaw_empiricalFreq_to_weight_ae`

rewrites the result in terms of the `empiricalFreq` abbreviation defined earlier.

Thus the current formal route is completely explicit. The mode of convergence is almost sure convergence. Integrability and identical distribution are discharged internally. The only non-trivial convergence hypothesis supplied by the caller is pairwise independence of the indicator random variables.

This exact statement should be reflected in the manuscript, rather than speaking only generically of a law-of-large-numbers application.

3.8 Separation of theorem layers

A useful conceptual decomposition of LF1 is the following four-layer split.

Layer 1: Measure-theoretic setup. This layer introduces the ontic measurable space, preparation region, deterministic flow, preparation measure, and outcome-region weights.

Layer 2: Trial-space construction. This layer introduces the repeated-trial sample space and the family of trial maps whose common law is the preparation measure.

Layer 3: Indicator observables and expectation bridge. This layer defines the indicator random variables and proves that their expectations agree with the real-valued ontic weights.

Layer 4: Frequency convergence. This layer applies the strong law of large numbers to obtain almost-sure convergence of empirical frequencies.

At the code level, the current repository refines this conceptual split slightly by isolating the expectation bridge into its own file, `Expectation.lean`, between `Indicators.lean` and `Convergence.lean`. This refinement is useful and should be reflected in the architectural description, because it cleanly separates indicator construction from the identification of expectations with ontic weights.

3.9 Repository structure

The public Lean4 implementation for LF1 is available at `https://github.com/zblore/csd-lean4`. The current repository layout mirrors the theorem architecture and is no longer merely prospective.

The current repository layout mirrors the theorem architecture and is no longer merely prospective. The LF1 development is exported from the top-level file `CsdLean4.lean`, which imports the following eight modules:

```
CsdLean4/LF1/Setup.lean
CsdLean4/LF1/Preparation.lean
CsdLean4/LF1/Outcomes.lean
CsdLean4/LF1/Trials.lean
CsdLean4/LF1/Indicators.lean
CsdLean4/LF1/Expectation.lean
CsdLean4/LF1/Convergence.lean
CsdLean4/LF1/MainTheorem.lean
```

Their roles are as follows.

- `Setup.lean`: ontic measurable space, finite measure, deterministic flow, preparation region.
- `Preparation.lean`: conditional preparation measure and the explicit rewriting formula `prepMeasure_apply`.
- `Outcomes.lean`: measurable ontic outcome regions, pullback events, preparation-side events, and ontic weights.
- `Trials.lean`: repeated-trial model on an external sample space, including the common-law condition for trial maps.
- `Indicators.lean`: indicator observables and empirical frequencies.
- `Expectation.lean`: expectation-to-weight bridge.
- `Convergence.lean`: strong-law application and almost-sure convergence lemmas.
- `MainTheorem.lean`: the exported LF1 theorem and its immediate corollaries.

This modular layout is one of the strengths of the present implementation. It cleanly isolates reusable theorem layers and provides a stable interface for later formalisation targets such as LF2 through LF5.

3.10 What is specific to CSD and what is not

One of the main interpretive virtues of the formalisation is that it makes the CSD-specific content easy to identify. Most of the theorem is standard probability theory. The specifically CSD-shaped inputs are comparatively few:

1. the interpretation of Σ as an ontic state space,
2. the use of μ_L as the physically relevant conserved measure,
3. the interpretation of Ω_0 as a preparation ensemble,
4. the interpretation of the Ω_i^Σ as outcome basins produced by a fixed experimental context,
5. the deterministic evolution map Φ_t.

Everything after these declarations is, in essence, a theorem of repeated-trial probability theory. This is not a weakness. On the contrary, it is exactly what the formalisation is meant to clarify. LF1 does not claim that the probability rule itself is unique. It claims that once a deterministic measure-preserving model and repeated preparation structure are given, stable frequencies follow from normalised volume weights. The Lean architecture makes that claim precise and no stronger.

4 Assumptions audit

4.1 Purpose of the audit

A central aim of LF1 is not only to formalise the repeated-trial frequency theorem, but also to make its logical dependencies completely explicit. In informal foundational arguments, mathematical facts, modelling choices, and physical assumptions are often presented together in compressed form. One contribution of machine verification is that this compression can no longer be maintained. Every object must be typed, every theorem must state its hypotheses, and every external dependency must be declared.

For that reason, the present section separates the ingredients of LF1 into three distinct classes:

1. statements proved internally in the Lean development,
2. results imported from Mathlib,
3. assumptions that belong to the physical model rather than to the mathematical proof.

This division is important both mathematically and conceptually. Mathematically, it clarifies the real content of the formal theorem. Conceptually, it prevents the foundational interpretation of LF1 from becoming stronger than the underlying formal result warrants.

4.2 Internally proved statements

The internal contribution of LF1 is now best described as an implemented theorem chain, not merely as a proposed architecture. The following statements are already proved within the current repository.

Normalised preparation measure. The development constructs the conditional preparation measure

$$\mu_{\mathrm{prep}}(A) = \frac{\mu_L(A \cap \Omega_0)}{\mu_L(\Omega_0)},$$

and provides the explicit rewriting lemma

`prepMeasure_apply`.

Measurable outcome-region events. For a fixed measurable outcome region O, the repository constructs the pulled-back event

$$O.\mathrm{preEvent} = \Phi^{-1}(O.\Omega)$$

and the preparation-side event

$$O.\mathrm{prepEvent} = \Omega_0 \cap O.\mathrm{preEvent},$$

and proves their measurability.

Ontic weight formula. The development defines the ontic weight $O.\mathrm{weight}$ through the preparation measure and proves the explicit identity

$$O.\mathrm{weight} = \frac{\mu_L(O.\mathrm{prepEvent})}{\mu_L(\Omega_0)}.$$

Trial-event and trial-probability infrastructure. For each repeated-trial model T, the repository defines `trialEvent` and `trialProb`, and proves

$$T.\mathrm{trialProb}(O, n) = O.\mathrm{weight}.$$

Expectation bridge. The key expectation lemmas already present are

`integral_indicatorRV_eq_trialProbReal` and `integral_indicatorRV_eq_weightReal`.

These show that the expectation of the indicator observable equals the corresponding real-valued ontic weight.

Strong-law convergence chain. The repository proves the successive convergence lemmas

`strongLaw_indicator_to_mean_ae`, `strongLaw_indicator_to_weight_ae`, `strongLaw_empiricalFreq_to_weigh`

Exported main theorem. The final theorem is exported as

`LF1_main_theorem_ae`,

with the theorem signature matching the manuscript-level statement: for a repeated-trial model T, a fixed outcome region O, and a pairwise-independence hypothesis, empirical frequencies converge almost surely to $O.\mathrm{weightReal}$.

This theorem chain is the genuine internal mathematical content of LF1 as it presently stands.

4.3 Imported mathematical results from Mathlib

LF1 is intentionally modest in its imported mathematical dependencies. The core external convergence result used in the present repository is the real-valued strong law of large numbers

`strong_law_ae_real`,

imported via `Mathlib.Probability.StrongLaw`.

The repository also imports standard infrastructure from Mathlib for identical distribution and repeated-trial measure handling, including the modules for `IdentDistrib` and product-measure constructions. These are used to organise the repeated-trial probability model and to discharge the standard probability-theoretic hypotheses required by the strong law.

It is important to state this precisely. The current implementation does not present the theorem as if it were proved from first principles inside LF1 alone. Rather, LF1 combines a machine-verified CSD-specific measure-and-event construction with a standard imported law-of-large-numbers theorem from Mathlib. This is exactly the correct methodological stance for the first Lean target in the programme.

4.4 Physical-model assumptions

The mathematical proof in LF1 rests on a small and explicit package of physical-model assumptions.

Finite ontic measure. The ontic measurable space carries a finite measure μ_L, interpreted physically as Liouville measure or its abstract measurable analogue.

Deterministic ontic flow. There is a deterministic evolution map

$$\Phi : \Sigma \to \Sigma,$$

interpreted physically as the ontic flow of the isolated system under the chosen experimental context.

Preparation region. There is a measurable preparation region $\Omega_0 \subseteq \Sigma$ of non-zero measure, allowing construction of the conditional preparation measure.

Outcome region. For the current theorem, one fixes a single measurable ontic outcome region $O.\Omega \subseteq \Sigma$. The wider physical interpretation may involve a family of such regions associated with a measurement context, but the machine-verified LF1 theorem is stated for one region at a time.

Repeated preparation. Repeated trials are modelled by measurable trial maps X_n whose common law is the preparation measure. At the final theorem level, the only non-trivial repeated-trial hypothesis supplied explicitly is pairwise independence of the associated indicator random variables.

Measure preservation. The ontic setup carries a measure-preservation hypothesis for Φ, reflecting Liouville invariance. In the present LF1 layer this enters directly only through the induced measurability of Φ, while its fuller dynamical role is reserved for later layers.

These assumptions are exactly the point of LF1. The paper does not claim to derive them from deeper symplectic or Hamiltonian structure. It treats them as the physically motivated ontic input package on top of which the repeated-trial frequency theorem is machine-verified.

4.5 Assumptions not made

An audit is clearer if it also records what the formalisation does *not* assume. The following ingredients are outside the scope of LF1 and are not needed for the main theorem.

No ergodicity assumption. LF1 does not assume that a single ontic orbit explores the full preparation region in time. The theorem is about repeated preparation, not about long-time averages on a single trajectory.

No stochastic ontic dynamics. The ontic evolution remains deterministic throughout. Probability enters only through the repeated-trial preparation model.

No projective-state-space structure. The theorem does not require $\mathbb{CP}^{N-1}$, Fubini-Study measure, or any projective geometric apparatus. Those belong to later measure-bridge and operational-weight layers.

No uniqueness theorem for Born-form weights. LF1 does not assume or prove any Gleason-type, Busch-type, or other uniqueness theorem for quantum probabilities. It is a frequency theorem only.

No mixed states or POVMs. The theorem is formulated entirely at the level of preparation regions and outcome events in the ontic space. Mixed-state structure, density operators, POVMs, and subsystem reduction belong to later formalisation stages.

This list is scientifically important. It shows that LF1 isolates a comparatively small theorem kernel. That is a virtue. A narrow theorem with a transparent hypothesis set is easier to verify, easier to criticise fairly, and easier to reuse.

4.6 Audit summary

The assumptions audit shows that LF1 has a deliberately narrow and transparent logical profile.

First, the CSD-specific content is confined to the ontic measurable setup: the finite measure μ_L, the deterministic flow Φ, the preparation region Ω_0, and the fixed measurable outcome region O.

Second, the repository already proves the full bridge from these ontic inputs to the repeated-trial theorem infrastructure: the conditional preparation measure, the measurable pulled-back event, the ontic weight formula, the indicator observables, and the expectation-to-weight identity.

Third, the final convergence step relies on a standard imported strong law of large numbers. The explicit non-trivial repeated-trial hypothesis supplied at that stage is pairwise independence of the indicator random variables. Integrability and identical distribution are obtained automatically from the earlier formal structure.

In that precise sense, LF1 does exactly what it should do. It makes the statistical backbone of deterministic volume typicality explicit, formal, and no stronger than the theorem actually warrants.

5 Relation to the CSD programme

5.1 Role of LF1 within the finite-dimensional programme

Within the Constraint-Surface Dynamics programme, LF1 formalises the first theorem in the statistical backbone of the finite-dimensional reconstruction. Its role is narrow but structurally important. Paper A establishes that, under an explicit repeated-trial preparation model, deterministic measure-preserving dynamics on a finite-measure ontic state space yield stable long-run outcome frequencies equal to the corresponding normalised volume ratios (Blore 2026a). LF1 is the machine-verified version of that claim.

This role should be stated carefully. LF1 does not introduce a new physical principle. Nor does it attempt to settle the wider interpretive or operational questions of the programme. Its contribution is methodological and foundational in a precise sense: it turns the determinism-to-frequency bridge into a theorem whose assumptions, imported dependencies, and formal proof structure are all fully explicit.

In the internal dependency logic of the CSD programme, this is the first formal layer because later claims presuppose it. Any account that aims to connect ontic volume to effective quantum weights, to mixed-state structure, or to controlled measurement dynamics still requires a theorem showing how repeated observed frequencies arise from deterministic geometry under repeated preparation. LF1 isolates exactly that theorem.

5.2 Relation to Paper A

LF1 should be read as a formal companion to Paper A rather than as a replacement for it. Paper A is the conceptual and mathematical source of the repeated-trial volume-typicality claim. It introduces the preparation region, the outcome

regions, the deterministic volume-preserving flow, and the repeated independent preparation model under which empirical frequencies converge to the corresponding measure ratios (Blore 2026a).

LF1 takes that theorem spine and expresses it in a theorem-prover-friendly form. In doing so, it performs three tasks that Paper A, by design, does not attempt in full detail.

First, it rewrites the theorem in the canonical notation adopted after TN0. The present paper therefore uses (Σ, μ_L, Φ_t) in place of the earlier notation of Paper A.

Second, it separates the repeated-trial sampling model from the deterministic dynamics themselves. This distinction is conceptually present in Paper A, but the formalisation forces it to be stated with complete precision.

Third, it converts the theorem into a reusable formal module. Later Lean papers in the branch are intended to build on this module directly rather than to restate the same probabilistic infrastructure each time.

Accordingly, LF1 should be cited as the machine-verified formalisation of the Paper A backbone theorem, not as an independent route to the same conclusion.

5.3 Why LF1 is not yet a Born-rule paper

The current paper must also state clearly what it does *not* accomplish. The theorem formalised in LF1 is a frequency theorem. It shows that, under repeated preparation from a conditional measure on an ontic state space, empirical frequencies converge to the corresponding normalised volume weights. This is the statistical backbone of a volume-based account of outcome regularities. It is not yet a derivation of the Born rule.

This distinction is already explicit in Paper A itself. Paper A states that its result is a compatibility statement: if the relevant outcome volumes align with squared-amplitude weights, then the resulting frequencies match the Born rule, but the paper does not itself explain why that alignment should hold (Blore 2026a). That further step belongs to later work.

For the same reason, LF1 does not establish the general finite-dimensional quantum probability rule. It contains neither the projective measure bridge nor the operational uniqueness layer. In the current programme logic, those belong to later results and later Lean papers. Thus LF1 should be presented as a machine-verified volume-typicality theorem and nothing stronger.

5.4 Relation to the later Lean papers

The Lean branch of the current paper index is organised so that LF1 supplies the formal kernel for later developments. In that branch, LF2 addresses the measure bridge and Born-weight wrapper, LF3 addresses mixed states, POVMs, and reduction, LF4 addresses control Hamiltonians and outcome basins, and LF5 addresses outcome-conditioned update and sequential circuits (Blore 2026e).

The dependence structure is conceptually straightforward.

LF2 requires a clean formal account of how repeated-trial frequencies are tied to normalised volume weights before one can ask how those weights are transported to projective state space.

LF3 requires a stable formal probability layer before one can treat density operators, subsystem marginals, and POVM probabilities in a systematic way.

LF4 requires the event-to-indicator and frequency machinery of LF1 in order to express gate-dependent basin transport as a measurable change in outcome statistics under deterministic control.

LF5 requires the same probabilistic infrastructure in order to describe sequential experiments, where a new preparation for the next stage is conditioned on a realised earlier outcome.

Thus LF1 should be understood as a foundational software and theorem layer for the rest of the Lean branch. Its theorem is simple compared with the later papers, but the later papers are cleaner precisely because LF1 isolates the repeated-trial kernel in advance.

5.5 Why LF1 is the correct first Lean target

The choice of LF1 as the first Lean formalisation target is not arbitrary. In the current dependency map of the programme, the recommended first internal Lean target is the typicality and frequency-to-volume link, because once the dynamical assumptions are declared, the result is largely measure-theoretic and therefore well matched to the existing theorem-proving ecosystem (Blore 2026f). Later in the same internal map, the second recommended Lean target is the

finite-dimensional affine-functional and marginalisation layer, with Gleason-class and Naimark-type results allowed as imported theorems at first.

This ordering has a clear methodological advantage. It allows the Lean branch to begin with a theorem that is both mathematically precise and realistically achievable. It also forces the programme to state its first probabilistic claim with maximal clarity before moving on to more elaborate structures such as projective-state-space geometry, operational weight functionals, or post-measurement update.

For that reason, LF1 should not be viewed as a preliminary exercise. It is the natural first theorem to formalise because it is the narrowest theorem on which several later layers depend.

5.6 Relation to the finite-dimensional closure package

In the broader CSD programme, the finite-dimensional reconstruction is treated as a layered structure rather than as a single theorem. The internal dependency map states that a sector-conditional finite-dimensional closure requires the backbone results, the operational completion provided by TN4, the sequential-update layer provided by TN6, and at least one constructive basin model together with a worked template example (Blore 2026f). LF1 addresses only the first of these components.

This is a strength rather than a weakness. By keeping LF1 narrow, the present paper makes it easier to distinguish what has already been formalised from what still belongs to later stages. In particular, LF1 does not claim to close the finite-dimensional programme by itself. It contributes one clearly identified theorem to that closure package.

The practical consequence is that LF1 strengthens the formal basis of the programme, but the programme should still describe its broader finite-dimensional claims as depending on later layers until those layers are also formalised.

5.7 Relation to applications and the QC path

The present paper also has a clear downstream role in the quantum-information and quantum-computation path of the programme. The current paper index places QI2, *Quantum Computation as Geometric Dynamics*, in the active application-facing branch, and places LF4 and LF5 later in the Lean branch as the formal counterparts of control-dependent basin transport and sequential circuit update (Blore 2026e).

LF1 does not by itself explain quantum gates or circuit dynamics in full. What it does provide is the first rigorous ingredient needed for such an explanation: once a control-dependent Hamiltonian and an outcome-basin structure are supplied, LF1 tells us how repeated observed frequencies are related to the corresponding normalised geometric weights. In that sense, LF1 is not the whole QC explanation, but it is part of the formal substrate on which the QC explanation is built.

This is exactly the right level of claim for the present paper. It contributes to the later QC-facing story without pretending to complete it.

5.8 Boundary of programme-level significance

It is useful to conclude this section by stating the programme-level significance of LF1 in one sentence. LF1 shows that the first statistical theorem in the finite-dimensional CSD architecture can be expressed as a transparent Lean4 theorem whose assumptions are explicit and whose proof no longer depends on informal compression. It does not yet provide the full probability rule, the full operational layer, or the full control-and-update story. Those remain the tasks of later papers.

That boundary should remain visible throughout. The value of LF1 lies precisely in the fact that it does one thing cleanly. It machine-checks the repeated-trial volume-typicality backbone and thereby stabilises the first formal rung of the wider CSD programme.

6 Discussion

6.1 What the formalisation clarifies

The present formalisation clarifies several points that are often left implicit in informal foundational discussions.

First, it separates deterministic ontic evolution from probabilistic repeated preparation. A single trial is completely deterministic at the ontic level. The probability space enters only to model repeated preparation of initial conditions.

Second, it identifies the exact measure-theoretic object whose value controls the frequency theorem: not an abstract probability postulate, but the normalised ontic weight of a pulled-back measurable outcome event.

Third, it shows that the expectation-to-weight bridge is itself a non-trivial formal layer and deserves explicit isolation. In the repository this appears as its own file, `Expectation.lean`, between the indicator construction and the convergence theorem.

Fourth, it reveals the exact theorem architecture of LF1 as a structured composition of measure restriction, repeated-trial sample-space modelling, deterministic event pullback, indicator observables, expectation identification, and a law-of-large-numbers step.

Finally, it clarifies the boundary of the result. LF1 is a theorem about repeated-trial frequency convergence for deterministic ontic models with a probabilistic preparation scheme. It is not yet a theorem about Born weights, projective geometry, mixed states, or post-measurement update. That sharper boundary is a strength, not a weakness, because it makes the role of LF1 within the wider CSD programme fully explicit.

6.2 What LF1 does not establish

The narrowness of LF1 should be regarded as a strength, but it is equally important to state clearly what remains outside its scope.

Most importantly, LF1 does not provide a machine-verified derivation of the Born rule. The theorem proved here yields long-run convergence of empirical frequencies to normalised ontic volume ratios under repeated preparation. It does not yet show why those weights must coincide with the standard quadratic quantum weights in full finite-dimensional generality. That further step requires additional structure beyond the present paper.

In particular, LF1 does not establish a measure bridge from ontic volume on Σ to the effective projective measure on $\mathbb{CP}^{N-1}$. Nor does it establish the operational uniqueness of the Born-form probability assignment on projective state space. Those tasks belong to later layers of the programme.

LF1 also does not classify the physically realised outcome partitions. The theorem assumes the existence of measurable outcome regions with the required null-boundary and partition properties, but it does not derive them from a full dynamical construction of the measuring apparatus. The question of which partitions are physically admissible remains a separate issue.

Similarly, LF1 does not treat mixed states, POVMs, subsystem reduction, control-dependent Hamiltonians, outcome-conditioned update, or sequential experiments. All of these require additional formal machinery. The present paper deliberately avoids them in order to isolate the cleanest theorem kernel.

Finally, LF1 does not claim that all aspects of the finite-dimensional CSD programme are now formally closed. It settles one theorem layer only. Its contribution is foundational in the architectural sense, not exhaustive in the programme-wide sense.

6.3 Methodological lessons from the Lean development

The formalisation also yields several methodological lessons that are useful beyond the immediate theorem.

The first lesson is that formal verification is especially valuable when a theorem sits at the boundary between mathematics and physical interpretation. In such cases, a proof that appears conceptually straightforward may still conceal unspoken assumptions. Lean does not permit such assumptions to remain tacit. This improves not only mathematical rigour but also interpretive hygiene.

The second lesson is that a good formalisation target is often narrower than the corresponding informal claim. The natural temptation in a foundational programme is to formalise large slogans such as “deterministic geometry yields quantum probabilities.” In practice, progress is more reliable when this slogan is decomposed into sharply delimited theorem layers. LF1 represents one such layer. It is substantial enough to matter, but narrow enough to be realistically machine-checked.

The third lesson is that the distinction between imported theorems and newly formalised theorem chains is scientifically productive rather than embarrassing. In a mature theorem-proving ecosystem, one should expect to inherit measure theory and probability infrastructure from the existing library. The intellectual task is to identify exactly where the new formal contribution begins. In LF1, that contribution begins when the physical-model data are assembled into the specific repeated-trial theorem chain relevant to the CSD setting.

The fourth lesson is that notation discipline matters. One practical benefit of working after TN0 is that the ontic measure, flow, and region notation are already stabilised. This reduces ambiguity and makes later cross-paper formal reuse more realistic. The formalisation process therefore supports the broader programme aim of keeping terminology and symbol use consistent across technical notes and papers.

6.4 Implications for the wider Lean branch

LF1 is best understood as the first reusable module in a larger formalisation programme. Its theorem kernel is expected to support later Lean papers in at least four ways.

First, later work on the measure bridge can reuse the preparation-measure and event-weight infrastructure developed here. Even when the relevant outcome weights are eventually pushed forward to projective state space, the repeated-trial frequency layer will remain unchanged.

Second, later work on mixed states, POVMs, and subsystem reduction can reuse the trial-space and expectation machinery of LF1. The objects become richer, but the underlying probability-theoretic spine remains similar.

Third, later work on control Hamiltonians and outcome basins can reuse the event-to-indicator pipeline of LF1. In that setting, the main novelty will not be the probability layer itself, but the control-dependent construction of the measurable events whose frequencies are to be tracked.

Fourth, later work on sequential update and circuits can reuse the same machinery once the post-measurement preparation for the next stage has been defined. In this sense, LF1 provides infrastructure not only for theorem statements but also for software architecture within the repository.

This reuse is one of the main reasons for keeping LF1 narrow. A theorem that does one thing cleanly is easier to import into later formal work than a paper that tries to formalise several layers at once.

6.5 Limitations and open questions

Although LF1 is intentionally narrow, it still leaves several important questions open.

One open question concerns the physical status of the repeated independent preparation model. In the present paper, it is treated as an explicit modelling assumption because that is the correct logical status for the theorem proved here. A deeper analysis of how such repeated preparation arises from the physical architecture of experimental protocols would belong to a later paper.

A second open question concerns the dynamical construction of outcome regions. LF1 assumes the measurable outcome basins are given. For foundational and application-facing purposes, however, it is desirable to explain how those basins arise from specific system-apparatus couplings. That problem belongs to the later control-Hamiltonian layer.

A third open question concerns the passage from ontic volume to effective projective quantum weights. The present paper does not address the projective geometry of quantum theory, and therefore does not yet connect the formal theorem directly to standard Born-form statistics.

A fourth open question concerns the extent to which later finite-dimensional layers can be formalised internally, rather than by importing external uniqueness theorems. LF1 does not resolve that issue, but it helps by isolating the part of the programme that is already fully formalisation-ready.

These limitations should not be read as objections to LF1. They are the natural boundaries of a paper designed to formalise one theorem layer well.

6.6 Concluding perspective

LF1 shows that the first statistical theorem in the finite-dimensional CSD architecture can be made fully explicit in a modern theorem prover. This is valuable for two reasons. At the technical level, it provides a machine-checked version of the repeated-trial volume-typicality backbone. At the conceptual level, it clarifies exactly what that backbone does and does not establish.

The main result is therefore modest but significant. Deterministic measure-preserving dynamics, together with a measurable preparation ensemble and an explicit repeated-trial sampling model, suffice to generate long-run frequencies equal to normalised volume weights. This does not yet complete the wider probability story of the programme. It does, however, stabilise the first formal rung of that story.

For a foundational research programme, that is already a meaningful achievement. A theorem whose assumptions are transparent, whose proof is machine checked, and whose scope is carefully bounded provides a firmer basis for later claims than an informal argument whose dependencies remain partially compressed. LF1 is intended to provide exactly that firmer basis.

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